# Unit 2: Matter – Short Notes

**Key Concept**

- When heat energy is added, the kinetic energy increases first, so the temperature rises.
- At the phase change (melting point or boiling point), the kinetic energy does not increase; the heat supplied increases the potential energy instead, so the temperature stays constant.
- When energy is released, the opposite happens: the kinetic energy decreases first, and at the phase change the potential energy decreases while the temperature stays constant.

## 2.1 Changes of States of Matter

A state change (phase transition) is the change of matter from one state to another (solid ↔ liquid ↔ gas, or solid ↔ gas directly).

- The temperature stays constant during the change, because the kinetic energy stays the same.
- The potential energy changes, because inter-particle forces are overcome or formed.

### 1. Melting (Solid → Liquid)

- The particles vibrate in fixed positions.
- The kinetic energy increases first. At the phase change, the kinetic energy does not increase; the heat supplied increases the potential energy.
- Energy is added, so the potential energy increases and the forces weaken.
- The temperature stays constant.

### 2. Boiling (Liquid → Gas)

- The particles are closely packed and move past each other.
- The kinetic energy increases first. At the phase change, the kinetic energy does not increase; the heat supplied increases the potential energy.
- Energy is added, so the potential energy increases and the attractive forces are overcome.
- The temperature stays constant at the boiling point.

### 3. Freezing (Liquid → Solid)

- The particles move around each other.
- The kinetic energy decreases first. At the phase change, the kinetic energy does not decrease; the energy released decreases the potential energy.
- Energy is released, so the potential energy decreases and the forces pull the particles into a fixed arrangement.
- The temperature stays constant.

### 4. Condensation (Gas → Liquid)

- The particles are far apart and move rapidly.
- The kinetic energy decreases first. At the phase change, the kinetic energy does not decrease; the energy released decreases the potential energy.

- Energy is released, so the potential energy decreases and the forces pull the particles closer to form a liquid.
- The temperature stays constant.

### 5. Sublimation (Solid → Gas)

- The solid turns directly into a gas without passing through the liquid phase.
- The particles are tightly packed in fixed positions.
- The kinetic energy increases first. At the phase change, the kinetic energy does not increase; the heat supplied increases the potential energy.
- Energy is added, so the potential energy increases greatly and the solid forces are overcome.
- The temperature stays constant.

### 6. Deposition (Gas → Solid)

- The gas turns directly into a solid without passing through the liquid phase.
- The particles are far apart and move freely.
- The kinetic energy decreases first. At the phase change, the kinetic energy does not decrease; the energy released decreases the potential energy.
- Energy is released, so the potential energy decreases greatly and the forces pull the particles into a solid arrangement.
- The temperature stays constant.

### Quick Summary

| Process | Energy | Potential Energy |
|---|---|---|
| Melting, Boiling, Sublimation | Absorbed (added) | Increases |
| Freezing, Condensation, Deposition | Released | Decreases |

## 2.2 Heating and Cooling Curves

### 2.2.1 Heating Curve (Figure 2.1)

- Solid phase (A to B): The temperature increases. The kinetic energy and potential energy increase, so the particles vibrate more vigorously.
- Melting (B to C): The temperature stays constant. Energy is added, so the potential energy increases and the forces weaken (latent heat of fusion). The kinetic energy stays constant.
- Liquid phase (C to D): The temperature increases. The kinetic energy and potential energy increase, so the particles move more rapidly.
- Boiling (D to E): The temperature stays constant. Energy is added, so the potential energy increases and the attractive forces are overcome (latent heat of vaporization). The kinetic energy stays constant.
- Gas phase (E onward): The temperature increases. The kinetic energy and potential energy increase, so the particles move even more rapidly.

### 2.2.2 Cooling Curve (Figure 2.2)

- Gas phase (higher temperature to E): The temperature decreases. The kinetic energy and potential energy decrease, so the particles move less rapidly.

- Condensation (E to D): The temperature stays constant. Energy is released, so the potential energy decreases and the forces pull the particles closer to form a liquid. The kinetic energy stays constant.
- Liquid phase (D to C): The temperature decreases. The kinetic energy and potential energy decrease, so the particles move more slowly.
- Freezing (C to B): The temperature stays constant. Energy is released, so the potential energy decreases and the forces pull the particles into a fixed arrangement. The kinetic energy stays constant.
- Solid phase (B to A and below): The temperature decreases. The kinetic energy and potential energy decrease, so the particles vibrate less.

### 2.2.3 Kinetic Theory Interpretation

- Where the temperature changes, the kinetic energy changes (increases in heating, decreases in cooling).
- Where the temperature stays constant, the energy is used to change the phase, so the potential energy changes while the kinetic energy stays constant.

## 2.3 Evaporation and Boiling

### Q1. What is evaporation? How does temperature affect it?

- Evaporation is the process in which molecules at the surface of a liquid gain enough kinetic energy to overcome intermolecular forces and escape into the gaseous phase.
- Only surface molecules take part, and it occurs at any temperature below the boiling point.
- At higher temperature, the rate of evaporation increases, because more molecules have enough kinetic energy to escape.

### Q2. Why is evaporation a cooling process?

- The energy for evaporation is absorbed from the surroundings.
- The molecules with high kinetic energy are the ones that escape, so the average kinetic energy of the remaining molecules decreases, and the liquid and its surroundings become cooler. This is why sweating cools the body.
- Evaporation is faster at higher temperature, because more molecules have enough energy to escape.

### Q3. Give examples of evaporation.

- Drying of clothes on a line: water evaporates from the surface of the wet cloth, and the clothes dry faster at higher temperature because more molecules have enough kinetic energy to escape.
- Evaporation of water from lakes and oceans is part of the water cycle; it also increases at higher temperature.
- Sweat evaporating from the skin: high-energy molecules leave, the average kinetic energy of the remaining molecules decreases, so the body is cooled.

### Q4. What is boiling? What is boiling point?

- Boiling is the process in which a liquid turns into vapour when it is heated to its boiling point.
- Vapour bubbles form within the whole liquid (not just at the surface), rise to the surface and burst, releasing gas.

- The boiling point is the specific temperature at which boiling occurs; it depends on the liquid and the external pressure. For water at sea level it is 100 °C (212 °F).

### Q5. Why does the temperature remain constant during boiling (phase change)?

- Continuous heat is needed to keep the liquid at its boiling point.
- At the phase change, the added energy does not increase the kinetic energy; it increases the potential energy by overcoming the attractive forces between the particles (latent heat of vaporization).
- Since the kinetic energy stays the same, the temperature stays constant until the whole liquid has changed into gas.

### Q6. Write the difference between evaporation and boiling.

- Evaporation is a gradual process that happens only at the surface, at any temperature below the boiling point; its rate increases at higher temperature, and it causes cooling.
- Boiling is a rapid, vigorous process that happens throughout the liquid, with bubbles and steam, only at the boiling point.
- During boiling the temperature stays constant, because the added energy increases the potential energy (phase change).

## 2.3.1 Effect of Pressure on Evaporation and Boiling

### Q7. How does external pressure affect the boiling point?

- Higher pressure increases the boiling point and lower pressure decreases it.
- The liquid changes to gas (phase change, at constant temperature) at that new boiling point.
- This is why water boils at a lower temperature at high altitudes, where the atmospheric pressure is low.

### Q8. How does lower pressure affect evaporation and boiling?

- At lower atmospheric pressure, less external pressure acts on the liquid surface, so molecules escape more easily and the rate of evaporation increases.
- Lower pressure also reduces the boiling point, so the liquid changes to gas at a lower temperature.
- For example, water evaporates more quickly at high altitudes than at sea level, and it boils below 100 °C.

### Q9. How does higher pressure affect evaporation and boiling? Give an example.

- At higher atmospheric pressure, the increased external pressure makes it harder for molecules to escape, so the rate of evaporation decreases.
- Higher pressure also increases the boiling point, so the liquid needs a higher temperature to change into gas.
- In a pressure cooker, the high pressure suppresses boiling until a higher temperature is reached, so water boils at a higher temperature and food cooks faster.

## 2.4 Kinetic Particle Theory and Gas Laws

### Q10. What are the four variables of a gas? What are gas laws?

- A gas sample is described by four variables: number of moles (n), volume (V), pressure (P) and temperature (T).
- The relationships that show the effect of one variable on another, keeping the other two constant, are called gas laws.

### Q11. State Boyle's law and explain it with kinetic particle theory.

- Statement: for a fixed amount of gas at constant temperature, the volume of a gas is inversely proportional to its pressure.
- Formula: $V \propto 1/P$, $V = k(1/P)$, or $PV = \text{constant}$.
- When volume decreases, particles have less space and collide with the container walls more often, so pressure increases. When volume increases, collisions are less frequent and pressure decreases.
- The temperature is constant, so the average kinetic energy and speed of the particles do not change; only the frequency of collisions changes.

### Q12. State Charles's law and explain it with kinetic particle theory.

- Statement: for a fixed amount of gas at constant pressure, the volume of a gas is directly proportional to its absolute temperature.
- Formula: $V \propto T$, $V = kT$, or $V/T = \text{constant}$.
- When temperature rises, the kinetic energy of the particles increases and they move faster. To keep the pressure constant, the volume must increase so that particles travel further between collisions with the walls.
- If temperature decreases, particles move slower and the volume must decrease. So the volume changes to balance the change in frequency and force of collisions.

### Q13. State Avogadro's law and explain it with kinetic particle theory.

- Statement: for a gas at constant temperature and pressure, the volume of the gas is directly proportional to the number of moles of gas.
- Formula: $V \propto n$, $V = kn$, or $V/n = \text{constant}$.
- When the number of particles (moles) increases, more particles collide with the walls. To keep the pressure constant, the volume must increase so that the number of collisions per unit area stays the same. If the number of particles decreases, the volume decreases.
- Temperature keeps the kinetic energy of each particle the same, and pressure stays constant by adjusting the volume.

## 2.5 Diffusion of Gases

### Q14. What is diffusion? Give an example.

- Diffusion is the spreading of gas particles from a region where they are concentrated to the surrounding space, because of their random motion and collisions.
- Example: when perfume is sprayed, its molecules are concentrated in the sprayer at first; then they collide and spread across the room, so the smell can be detected even at a distance from the source.

## 2.5.1 Diffusion Process Explained by Kinetic Particle Theory

### Q15. Explain the diffusion process with the help of kinetic particle theory.

- When two different gases come into contact, their particles begin to move together. Because of their kinetic energy, each type of gas particle moves randomly in all directions.
- This random motion ensures that the particles from each of the two gases move into the other gas's space. This random and constant motion causes the gases to spread and mix up.
- The gas particles change direction as they move and collide with each other; as they change direction, they spread out and move through the space available to them.
- Eventually, the concentration of both types of gas particles is uniform across the entire space, and equilibrium is achieved.

## 2.5.2 Factors Affecting Diffusion

### Q16. What is the rate of diffusion? Which factors affect it?

- The rate of diffusion is the process by which molecules spread from areas of high concentration.
- It is affected by several factors, including molecular weight and temperature.

### Q17. State Graham's law of diffusion. How does molecular weight affect the rate of diffusion?

- Graham's law: the rate of diffusion is inversely proportional to the square root of the molar mass of the gas particles: $r \propto 1/\sqrt{m}$ at constant T and P. This means lighter molecules diffuse faster than heavier ones.
- According to kinetic particle theory, all molecules at the same temperature have the same average kinetic energy.
- Since $K.E = \frac{1}{2}mv^2$, lighter molecules (smaller m) have a greater velocity (v) than heavier molecules with the same kinetic energy.
- Because lighter molecules move faster, they can travel greater distances in a given time, resulting in a higher rate of diffusion.

### Q18. How does temperature affect the rate of diffusion?

- Diffusion rate increases with temperature. Higher temperatures give molecules more energy, which increases their kinetic energy.
- Since kinetic energy is proportional to the square of the velocity, a higher average kinetic energy results in a higher molecular velocity.
- Higher molecular velocities cause more frequent and energetic collisions between molecules, facilitating faster diffusion.

## Concept Assessment Exercise 2.1

### Q19. Explain why: (a) When you apply perfume to your palm, it makes you feel cold.

- The perfume evaporates, and the energy for evaporation is absorbed from the surroundings (the palm).
- The molecules with high kinetic energy are the ones that escape, so the average kinetic energy of the remaining molecules decreases, and the palm feels cold.

### Q20. Explain why: (b) We all sweat. As our sweat evaporates, it cools us.

- The energy for evaporation of sweat is absorbed from the surroundings (the body).
- The molecules with high kinetic energy are the ones that escape, so the average kinetic energy of the remaining molecules decreases, and the body is cooled.

## 2.6 Sublimation

### Q21. What is sublimation? Explain it with kinetic molecular theory.

- Sublimation is the process in which molecules of a substance in the solid state go to the vapour state directly, without going through the liquid state.
- According to the kinetic molecular theory, molecules are in constant random motion even below the melting point, and their average kinetic energy is directly proportional to the absolute temperature.
- Below the melting point, molecules are kept in the solid state by relatively weak intermolecular forces (compared to the covalent bonds within the molecule), so sometimes a molecule will have enough energy to break those bonds and go into the vapour state.
- Carbon dioxide, camphor, and naphthalene easily undergo sublimation.

### Q22. Give applications of sublimation in our life.

- Solid air fresheners: the solid fragrance slowly transforms into vapour, filling the entire space with an aroma.
- Textile industry: dye sublimation printers are used for printing at a reduced cost on objects such as T-shirts, pens, coffee mugs, and bags.
- Food industry: freeze-drying food items for preservation for a longer duration is due to sublimation.

### Q23. Give simple examples of deposition in daily life.

- The formation process of snow from water vapour.
- Black soot inside a vehicle exhaust.
- The making process of dry ice using carbon dioxide.

## 2.6.1 Importance of Diffusion Rates in Medicine

### Q24. Why are diffusion rates important in medicine?

- Kinetic particle theory helps to understand drug distribution rates. When you take a drug, by mouth or injection, it needs to diffuse from the injection site into your bloodstream.
- The diffusion rate determines how quickly a drug reaches its target area; drug molecules spread through different tissues and organs to reach your body's cells.
- Higher levels of diffusion usually mean your drug will work faster, e.g. a fast-acting pain reliever needs to dissolve quickly to provide immediate pain relief.

### Q25. Which factors affect the drug diffusion rate?

- Concentration gradient: if your drug is at a high concentration and your surrounding tissues are at a low concentration, the drug diffusion rate will be faster.
- Composition of the drug (liquid, gel, tablet): liquid drugs move more quickly than solid drugs, as their particles have already dispersed.
- Controlled-release formulations are used to control the rate of drug diffusion so that the therapeutic effect remains consistent over time.

## Exercise Short Questions

### i. Explain why the temperature remains constant during the phase change from ice to water.

- Continuous heat is needed to keep the ice at its melting point.
- At the phase change, the added energy does not increase the kinetic energy; it increases the potential energy and the forces weaken (latent heat of fusion).
- Since the kinetic energy stays the same, the temperature stays constant until the whole ice has changed into water.

### ii. What is the name given to the phase change when a solid is converted directly to a gas?

- Sublimation: the solid turns directly into a gas without passing through the liquid phase.

### iii. What is the difference between evaporation and boiling?

- Evaporation is a gradual process that happens only at the surface, at any temperature below the boiling point; its rate increases at higher temperature, and it causes cooling.
- Boiling is a rapid, vigorous process that happens throughout the liquid, with bubbles and steam, only at the boiling point.
- During boiling the temperature stays constant, because the added energy increases the potential energy (phase change).

### iv. Interpret a heating curve for water, identifying key phase changes.

- Solid phase (A to B), liquid phase (C to D) and gas phase (E onward): the temperature increases, because the kinetic energy increases.
- Melting (B to C): the temperature stays constant. Energy is added, so the potential energy increases and the forces weaken (latent heat of fusion).
- Boiling (D to E): the temperature stays constant. Energy is added, so the potential energy increases and the attractive forces are overcome (latent heat of vaporization).
- Where the temperature stays constant, the energy is used to change the phase, so the kinetic energy stays constant.

### v. Explain the effect of increasing temperature on the pressure of a gas in a sealed container.

- When temperature rises, the kinetic energy of the particles increases and they move faster.
- In a sealed container the volume cannot increase, so the particles collide with the container walls more often and with more force.
- So the pressure of the gas increases.

### vi. How does increasing the external pressure affect the boiling point of a liquid?

- Higher pressure increases the boiling point, so the liquid needs a higher temperature to change into gas.
- The increased external pressure makes it harder for molecules to escape.
- In a pressure cooker, the high pressure suppresses boiling until a higher temperature is reached, so water boils at a higher temperature and food cooks faster.

### vii. Describe how molecular mass influences the rate of diffusion.

- According to Graham's law, the rate of diffusion is inversely proportional to the square root of the molar mass of the gas particles ($r \propto 1/\sqrt{m}$ at constant T and P).
- All molecules at the same temperature have the same average kinetic energy. Since K.E = $\frac{1}{2}mv^2$, lighter molecules (smaller m) have a greater velocity.
- Because lighter molecules move faster, they travel greater distances in a given time, resulting in a higher rate of diffusion.

### viii. Give an example of sublimation and explain its practical application.

- Example: carbon dioxide (dry ice), camphor, and naphthalene easily undergo sublimation.
- Application: in solid air fresheners, the solid fragrance slowly transforms into vapour, filling the entire space with an aroma.
- In the food industry, freeze-drying food items for preservation for a longer duration is due to sublimation.

### ix. Why is the diffusion of gases faster at higher temperatures?

- Higher temperatures give molecules more energy, which increases their average kinetic energy.
- Since kinetic energy is proportional to the square of the velocity, this results in a higher molecular velocity.
- Higher molecular velocities cause more frequent and energetic collisions between molecules, facilitating faster diffusion.

### x. How does Avogadro's law relate to the volume and number of moles of gas?

- Statement: for a gas at constant temperature and pressure, the volume of the gas is directly proportional to the number of moles of gas.
- Formula: $V \propto n$, $V = kn$, or $V/n = \text{constant}$.
- When the number of particles (moles) increases, more particles collide with the walls; to keep the pressure constant, the volume must increase. If the number of particles decreases, the volume decreases.